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MODELLING OF CHLORINE DECAY IN MUNICIPAL WATER SUPPLIES

FANG HUA[1], J. R. WEST[1], R. A. BARKER[2] and C. F. FORSTER[1]*
[1]Department of Civil Engineering, University of Birmingham, Edgbaston, Birmingham B15 2TT, U.K. and [2]Severn Trent Water Ltd., Raynesway, Derby DE21 7JA, U.K.

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Abstract—This paper describes the effects which temperature and the initial chlorine concentration have on the chlorine decay in different water samples. It also reports the derivation of empirical formulae which describe these effects. The data show that chlorine decays more rapidly in fresh samples than in those which had been re-chlorinated with sodium hypochlorite to give a concentration of about 0.5 mg/l. The decay constants were found to be inversely proportional to the initial chlorine concentration in the former case and to the concentration added in the latter case. A combined first and second-order model is described which describes the decay more precisely than the conventional first-order model. Specifically, it shows that the chlorine decay is influenced by its initial concentration and explains why chlorine decays more rapidly in the initial stages. It also explains why the decay constant in the first-order model varies with the initial chlorine concentration. © 1999 Elsevier Science Ltd. All rights reserved

Key words—chlorine decay, temperature, chlorine concentration, water quality, decay equation

NOMENCLATURE

$A =$ constant, Arrhenius reaction opportunity
$C =$ chlorine concentration (mg l^{-1})
$C_0 =$ initial chlorine concentration (mg l^{-1})
$E =$ Arrhenius activation energy (l atm g-mol^{-1})
$k =$ total decay constant (h^{-1})
$k_b =$ bulk decay constant in the first-order model (h^{-1})
$k_w =$ wall decay constant in the first-order model (h^{-1})
$k_o =$ overall decay constant (l mg^{-1} h^{-1})
$k_1 =$ βk_o, decay constant in the combined first and second-order model (h^{-1})
$k_2 =$ αk_o, decay constant in the combined first and second-order model (l mg^{-1} h^{-1})
$R =$ gas constant (l atm g-mol^{-1} K^{-1})
$r^2 =$ coefficient of determination
$T =$ temperature (K or °C)
$t =$ time (h)
$[X] =$ total concentration of reactants (mg l^{-1})
$[X_0] =$ initial concentration of reactants (mg l^{-1})
$[X_i] =$ concentration of the ith reactant (mg l^{-1}) ($i = 1, 2, 3, \ldots, n$)
$\alpha =$ stoichiometry constant
$\beta =$ $[X_0] - \alpha C_0$
$\lambda_i =$ reaction rate constant of the ith reactant ($i = 1, 2, 3, \ldots, n$)

*Author to whom all correspondence should be addressed. [Tel.: +44-121-4145069; fax: +44-121-4145051; e-mail: c.f.forster@bham.ac.uk].

INTRODUCTION

Chlorination is one of the most widely used methods of disinfection. It is comparatively cheap, easy to use, effective in killing bacteria and has the added advantage that it remains within the system for a considerable time. However, because it has an ability to produce odours which consumers find easy to recognise, the levels of chlorine and particularly the variability of these levels, is one of the most frequent causes of customer complaint. In order to achieve a balance between sufficient chlorination to ensure bacteriology quality and, at the same time, providing customers with water which they find pleasant to drink, it is necessary to understand the mechanism of chlorine decay in water distribution systems and the factors affecting it.

Chlorine disappears due to its reactions with ammonia and organic compounds naturally present in source waters. However, the composition of these compounds in most cases remains unknown and their rates of reaction with chlorine are little known. In the water distribution system chlorine also reacts with the pipe wall. For these reasons, most kinetic models describing chlorine decay in natural water have been established empirically or semi-empirically.

The simplest model for chlorine decay is the first-order decay model in which the chlorine concentration is assumed to decay exponentially (Jadas-Hécart *et al.*, 1992; Zhang *et al.*, 1992; Rossman *et al.*, 1994) and for a given initial concentration and temperature, the first-order model can provide a fair approximation. The key issue of using the first-order decay equation is the determination of the decay constant and many experiments carried out in both laboratories and field studies have shown that it can vary with the quality of the source water, the water temperature, the Reynolds number and the material properties of the water pipes. Therefore, the total decay constant (k) is often expressed as the decay due to the chlorine demand of the pipe, known as the wall decay (k_w) and that due to the quality of the water itself, known as the bulk decay (k_b) (Biswas and Clark, 1993; Clark *et al.*, 1994; Beatty *et al.*, 1996) so that:

$$k = k_w + k_b \quad (1)$$

and it has been shown that both the wall and bulk decays are dependent on the water temperature (Beatty *et al.*, 1996).

Another aspect which must be considered is the fact that, because the model for either the bulk or the wall decay consists of only one parameter (the decay constant k_b or k_w) it fails to present the feature that chlorine decays relatively more rapidly, initially, than the best fit first-order curve as observed in many experiments. It has also been found that the value of the decay constant depends on the initial chlorine concentration (Dharmarajah and Patania, 1991; Zhang *et al.*, 1992). In order to incorporate these experimentally observed features into the kinetics, Noack and Doerr (1978), Qualls and Johnson (1983) and Dharmarajah and Patania (1991) have suggested a kinetic model in which two separated decay formulae were used, one was defined for the first few hours and the other for the remaining time. Because the initial decay is rapid, they suggested a second-order decay equation for the first phase and a first-order decay equation for the second phase. Ventresque *et al.* (1990), Jadas-Hécart *et al.* (1992) and Dossier-Berne *et al.* (1997) even suggested the use of a multiple parameter decay equation for the second phase in order to describe the chlorine decay behaviour more accurately.

This paper describes the effects of water temperature and the initial concentration of chlorine on the bulk decay of free chlorine together with the derivation of empirical formulae linking the bulk decay to the water temperature and initial chlorine concentration. It also describes the derivation of a semi-empirical chlorine decay equation based on the overall reaction stoichiometry. It has two independent parameters, each of which has clear physical meaning.

MATERIALS AND METHODS

Bulk decay may be isolated from wall decay by carrying out chlorine decay experiments on the source water under controlled conditions in the laboratory. In order to investigate the effects of water temperature on the bulk decay, experiments were carried out at different temperatures.

Two different water samples were used for the tests, one was taken from the final water sampling point at Frankley Water Treatment Works operated by Severn Trent Water (final water) and the other from a tap at the University of Birmingham (tap water) which is served by Frankley Water Treatment Works. The samples were examined at a range of temperatures; 6, 9, 12, 15, 18 and 20°C. With the tap water samples, two trials were carried out at each temperature, one used the water as sampled from the tap without re-chlorination (referred to as ambient case) and in the other the water was rechlorinated with sodium hypochlorite to an initial concentration of about 0.5 mg l^{-1}. Because of the limitations of the equipment, tests had to be carried out on different days for different temperatures. To minimise the effects of variations in the water quality, repeat tests were carried out weekly and repeated for one temperature from which the effects of water quality could be examined.

Before beginning any sampling, all the containers being used were cleaned to ensure that no chlorine demand was present. The cleaning involved washing them in distilled water, standing them overnight filled with distilled water which had been superchlorinated (10 mg l^{-1}), flushing them with distilled water and then allowing them to dry.

The chlorine decay in the samples (250 ml) was determined by incubating them at the fixed temperatures in dark bottles. The number of bottles prepared was based on the length of the trial, with one bottle being used for each measurement. The free chlorine concentrations were measured on duplicate sub-samples (10 ml) by adding the *N*,*N*-diethyl-*p*-phenylenediamine (DPD) tablets (Camlab, Cambridge). The colour which developed was measured in Hach pocket colorimeters (Camlab, Cambridge). The accuracy of the colorimeters was checked regularly against the standards produced by the manufacture and were found to be within ± 0.02 mg l^{-1}.

Three different samples were used to examine the kinetics of chlorine decay; one was taken from the final water sampling point at Frankley Water Treatment Works, one from a tap at the University of Birmingham and one was a solution of humic acid (Aldrich) in distilled water. Humic acid was used because it is one of the compounds which will contribute to the total organic carbon (TOC) in potable water. The chlorine decay of the potable water samples was measured at 9°C and the samples were not re-chlorinated. The distilled water containing humic acid was chlorinated to give initial concentration ratios of humic acid to chlorine which ranged from 1:1 to 4:1 and the decay was measured at 20°C.

RESULTS AND DISCUSSION

Figure 1 shows the free chlorine decay profiles of the tap water measured at 12°C. This was typical of all the temperatures investigated. These profiles confirm that for most of the time, the chlorine decay can fairly be represented by the following first-order decay equation:

$$C = C_0 e^{-k_b t} \quad (2)$$

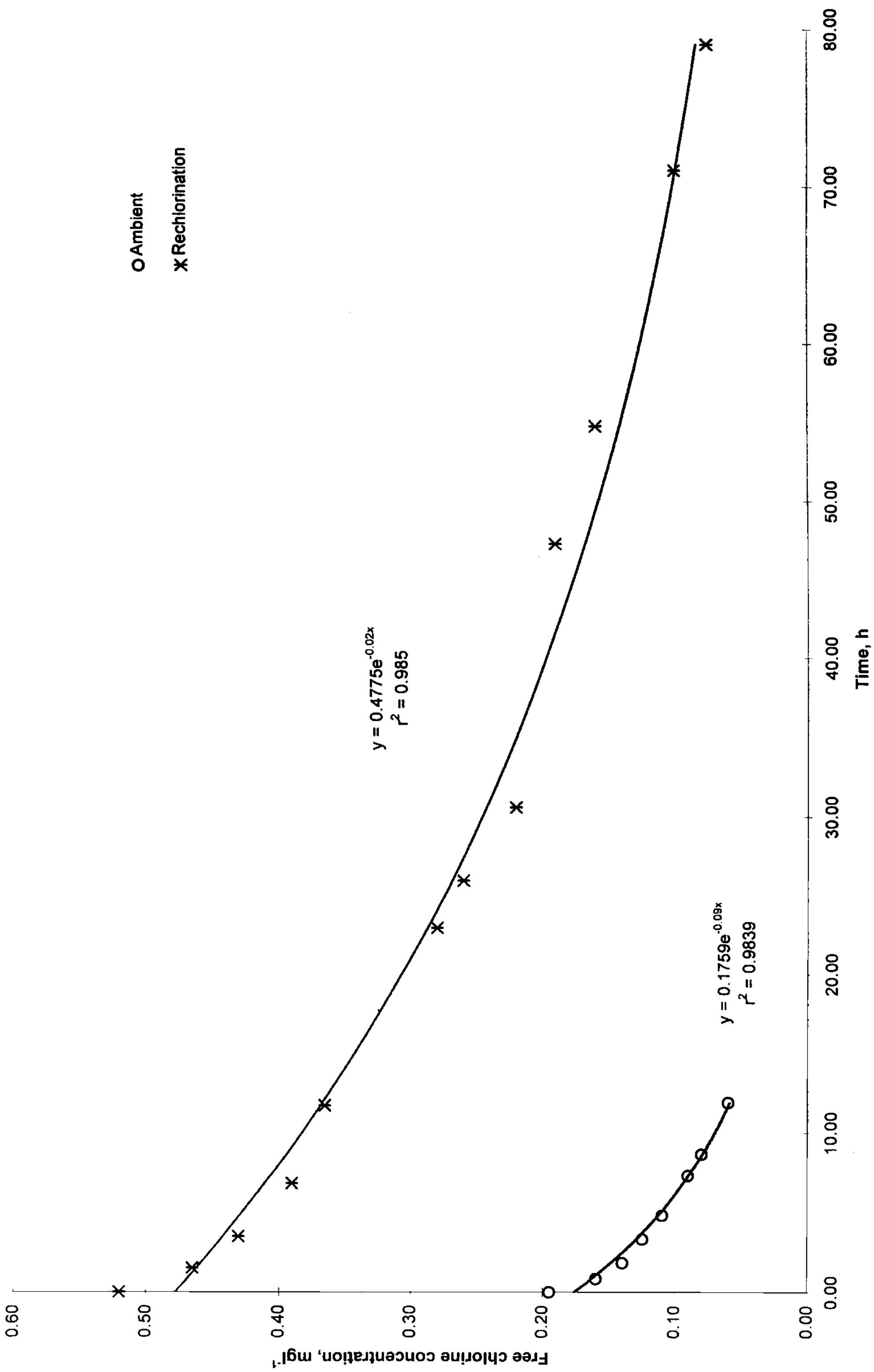

Fig. 1. Typical decay profiles for tap water at 12°C.

Table 1. Chlorine decay constants of tap water at different temperatures

Temperature (°C)	Initial chlorine concentration (mg l^{-1})		Bulk decay constant (h^{-1})	
	ambient	re-chlorination	ambient	re-chlorination
6	0.12	0.5	0.11	0.01
9	0.12	0.5	0.13	0.02
12	0.20	0.52	0.09	0.02
15	0.16	0.48	0.14	0.03
18	0.16	0.46	0.14	0.05
20	0.15	0.44	0.19	0.06
9(6)[a]	0.12	0.50	0.13	0.02
9(12)[a]	0.20	0.52	0.06	0.02
9(15)[a]	0.16	0.48	0.08	0.02
9(18/20)[a]	0.11	0.46	0.15	0.02

[a]The samples at 9°C were measured at the same time as the temperature in parentheses.

The decay constants for each temperature are given in Table 1 and were calculated using the least squares method to fit the experimental data with the regression line. In all cases, the coefficients of determination (r^2) were better than 0.93 with levels of significance of 0.001.

It can be seen that in the ambient samples, chlorine decays more rapidly than in the rechlorination case. The decay constant in the former case was found to be about 3–10 times greater than that found in the re-chlorination case. The general trend is that the higher the temperature, the higher the decay constant.

Table 1 also shows decay constants for the tap water at 9°C, which were measured on the same days as those described earlier. Because the temperature was constant in this case, the difference between these decay profiles reflects the effects of water quality. It was interesting to find that, for the ambient case, lower initial concentrations gave larger decay constants. The relationship between the initial concentration and the bulk decay constant for a fixed temperature for the three types of water is shown in Fig. 2 and, within the experimental limits, can approximately be represented by:

$$k_b \approx \frac{0.018}{C_0} - 0.024 \quad (3)$$

The chlorine decay constants in the water sampled at the treatment works (final water) were of a similar magnitude to those in the re-chlorinated tap water (Tables 1 and 2). This is, perhaps, not surprising as both samples represent recently chlorinated water, whereas the "ambient" tap water samples express the behaviour of water which has been reacting with chlorine for some time. The chlorine concentrations in the final water samples did vary from day to day and it was found that the decay constant was inversely proportional to the initial concentration (Fig. 2).

It is well-known that the rates of most chemical reactions increase with temperature. The change in rate constant with temperature can be expressed mathematically by the Arrhenius equation (Sawyer, 1994):

$$k_b = A e^{-(E/RT)} \quad (4)$$

This implies that a semi-logarithmic plot ($\ln k_b$ vs $1/T$) would be linear. However, the data in Tables 1 and 2 did not comply very well with this relationship. The data in Fig. 2 have demonstrated the involvement of the initial chlorine concentration. Therefore, relationships involving both the bulk decay constant and the initial chlorine concentration were investigated. Figure 3 shows a comparison of the experimental data in which $k_b C_0$ was used instead of k_b. As these were empirical relationships, for convenience, °C were used as the unit of temperature.

It can be seen that the empirical equations given below fit the experimental data, within the experimental limits, very well.

$$k_b C_0 = 0.0098 e^{0.0512T} \quad \text{for the ambient case} \quad (5)$$

$$k_b C_0 = 0.0029 e^{0.112T} \quad \text{for the rechlorination case} \quad (6)$$

$$k_b C_0 = 0.0050 e^{0.0841T} \quad \text{for the final water} \quad (7)$$

Table 2. Statistical parameters relating to the measurement of the bulk decay constant (k_b) of final water at different temperatures

Temperature (°C)	Initial chlorine concentrations (mg l^{-1})	k_b (h^{-1})	r^2	n
6	0.40	0.02	0.979	12
9	0.35	0.03	0.943	10
12	0.34	0.04	0.961	10
15	0.34	0.06	0.988	10
18	0.30	0.07	0.919	7
20	0.29	0.09	0.973	7

n is the number of data sets in each trial.

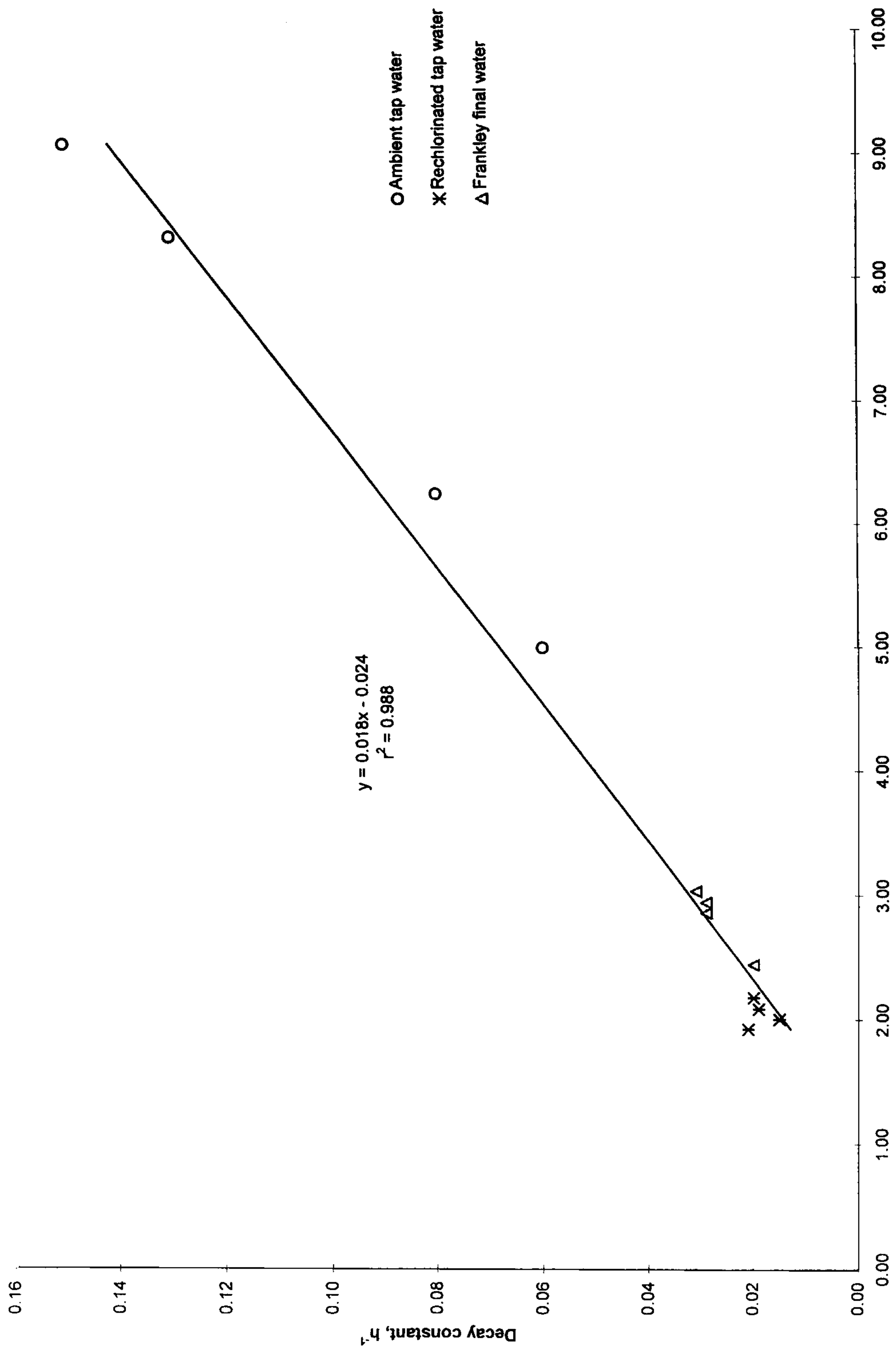

Fig. 2. The relationship between the decay rate (measured at 9°C) and the initial chlorine concentration.

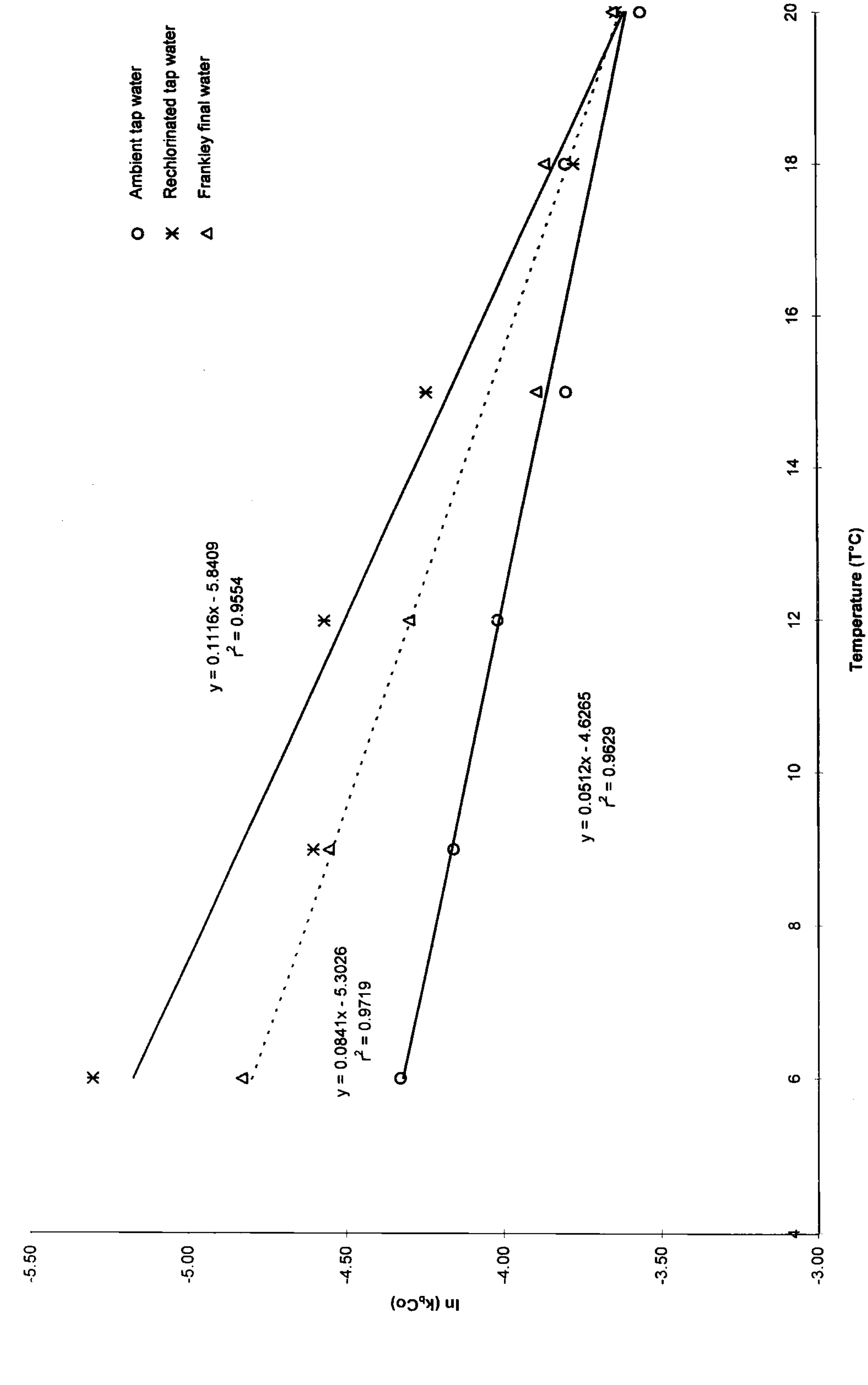

Fig. 3. Relationships between the decay constants and temperature.

Table 3. Results for distilled water containing humic acid (X)

Initial concentration ratio, $[X_0]/C_0$	C_0 (mg l^{-1})	Present model			First-order model	
		k_1 (h^{-1})	k_2 (l mg^{-1} h^{-1})	r^2	k_b (h^{-1})	r^2
1.04	0.46	−0.135	0.990	0.978	0.112	0.931
1.85	0.46	−0.130	0.900	0.980	0.120	0.970
2.78	0.36	0.110	0.921	0.990	0.230	0.990
4.03	0.49	0.048	1.130	0.989	0.224	0.945

A comparison of these equations shows that waters which have different histories have different relationships. Having incorporated temperature and the initial chlorine concentration, the differences must be associated with the concentration of organics in the water and the reactivity of these organics.

One way of incorporating the concentrations of organic compounds is to use the decay model whose derivation is given in Appendix A.

Equation (A.5) can be written as:

$$\frac{dC}{dt} = -k_1 C - k_2 C^2 \tag{8}$$

In a real water system, the chemical composition of the reactants and their initial concentrations are not known and are probably not measurable. Therefore, k_1 and k_2 have to be determined experimentally. This involves analyses of chlorine concentration/time sets. The analysis may be algebraic or by integration. Integrating equation (8) yields:

$$\frac{1}{C} + \frac{k_2}{k_1} = \left(\frac{1}{C_0} + \frac{k_2}{k_1}\right) e^{k_1 t} \tag{9}$$

k_1 and k_2 were determined by deriving the best fit of equation (9) with the experimental data. Iteration or a multi-parameter fitting programme may be used to achieve this. In this present study, the latter method was used.

Figure 4 shows the chlorine decay at 20°C of a sample of distilled water containing humic acid (Aldrich) with an initial concentration ratio of humic acid to chlorine of about 4:1. In order to compare the accuracy of the present model with the conventional first-order model, the decay constant k_b in the first-order model, k_1 and k_2 in the present model together with calculated coefficients of determination of the regression (r^2) are also shown in Fig. 4. It can be seen from these calculated coefficients that the combined first and second-order model provides better prediction than the first-order model.

Similar results were obtained for different initial concentration ratios of humic acid to chlorine (Table 3). As might be expected, the decay constants, k_2, for the different ratio samples, are similar (mean = 0.99 ± 0.09). This is because all samples contain only one reactant (i.e. humic acid) which should have the same reaction rate constant (k_o) and the same stoichiometry constant (α). The decay constants, k_1, are different because k_1 is related to the initial concentrations of both chlorine and reactant. As $k_1 = \beta k_o = ([X_0] - \alpha C_0) k_o = k_o C_0 ([X_0]/C_0 - \alpha)$, a small ratio of $[X_0]$ to C_0 results in a negative value for k_1. From the present study, using distilled water and a given concentration of humic acid, it was found that the stoichiometry constant for humic acid was in a range: $2 < \alpha < 3$ and that the overall reaction rate constant was in the range: $0.3 < k_o < 0.6$ (l mg^{-1} h^{-1}).

Figure 5 shows the comparisons of the first-order model and the present model using experimental data for a final water sample at a fixed temperature 9°C, taken from Frankley Water Treatment Works. Again, it can be seen that the present model is better than the first-order model. More results for the same water, but taken on different days are shown in Table 4. The treated final waters will contain differing concentrations of TOC and this is considered to be contributory to the cause of the differences in the decay constants k_2. In addition, the decay constants, k_1, in most of the samples, were found to be negative. These negative values imply that the initial chlorine concentration was higher than that required by the reactants, or in other words, the reactants were not present at a concentration sufficient to consume all the existing chlorine completely, a low $[X_0]$ to C_0 ratio.

Figure 6 shows the results of the tap water at the University of Birmingham, whose source was

Table 4. Results for final water samples

Sampling date	C_0 (mg l^{-1})	Present model			First-order model	
		k_1 (h^{-1})	k_2 (l mg^{-1} h^{-1})	r^2	k_b (h^{-1})	r^2
07/05/97	0.35	−0.004	0.199	0.992	0.029	0.926
12/05/97	0.41	−0.001	0.202	0.989	0.023	0.975
14/05/97	0.33	−0.001	0.183	0.981	0.031	0.943
19/05/97	0.34	0.000	0.108	0.983	0.029	0.964

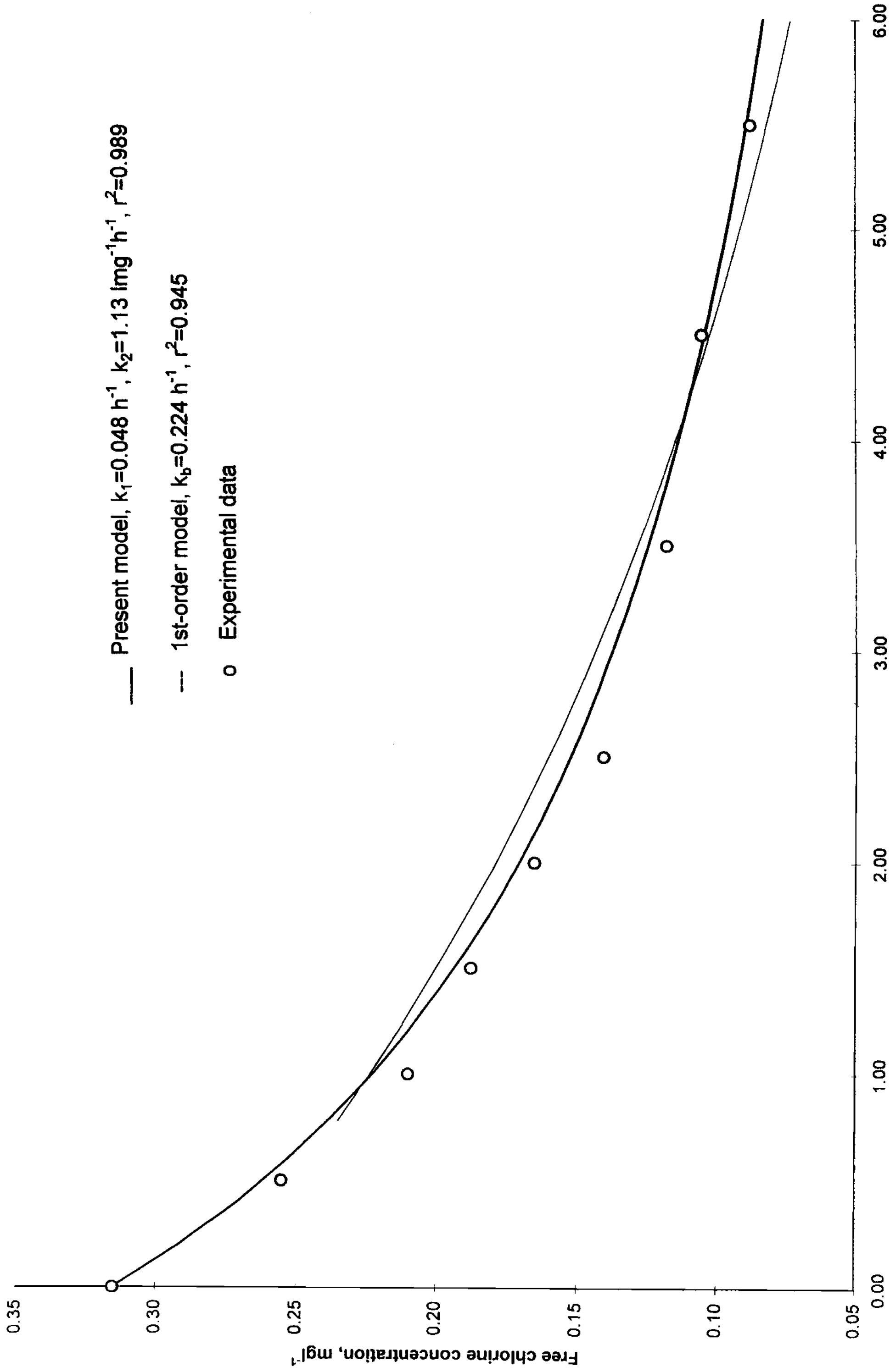

Fig. 4. Comparisons of model predictions and experimental data for the humic acid–distilled water system.

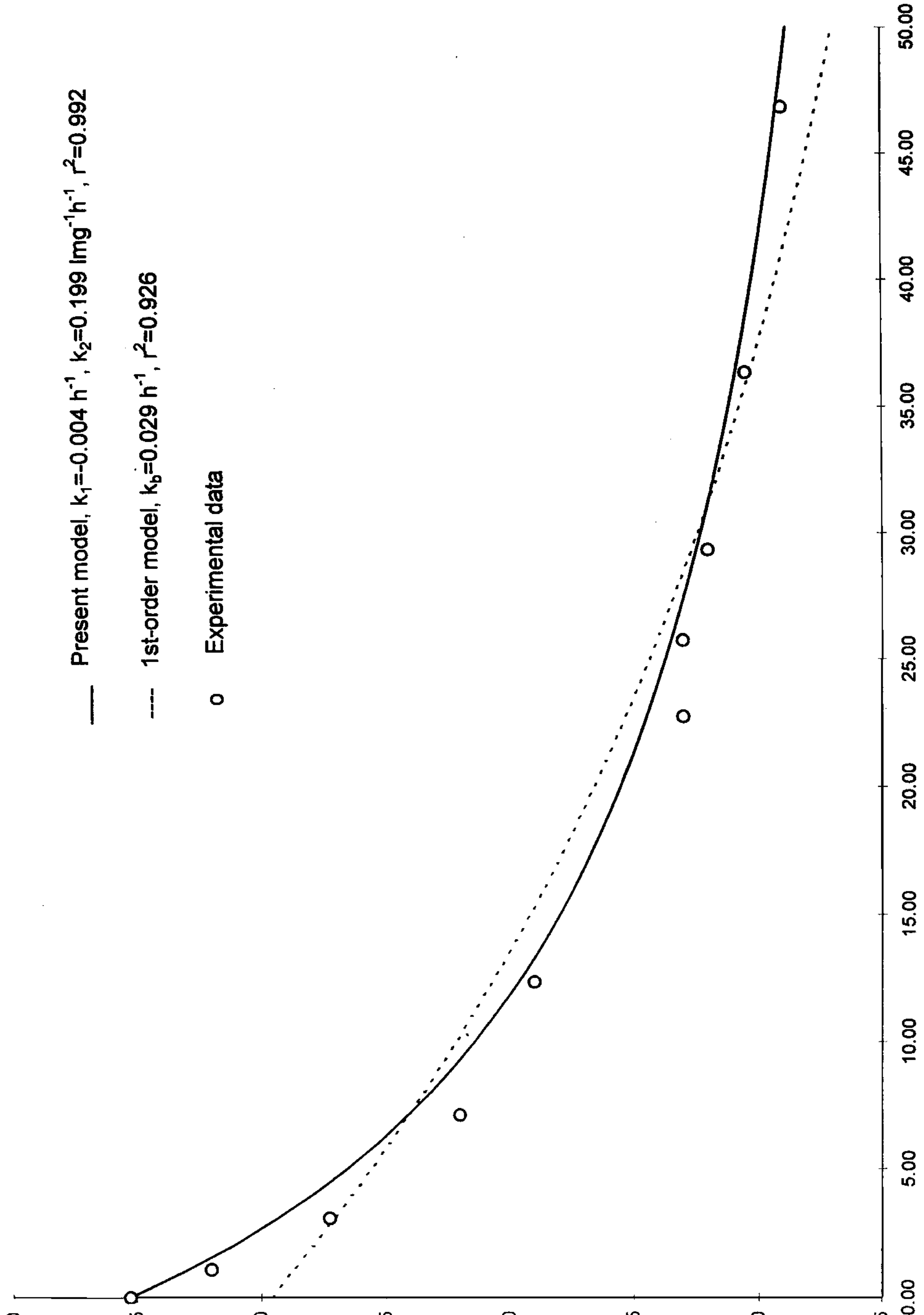

Fig. 5. Comparisons of model predictions and experimental data for Frankley final water.

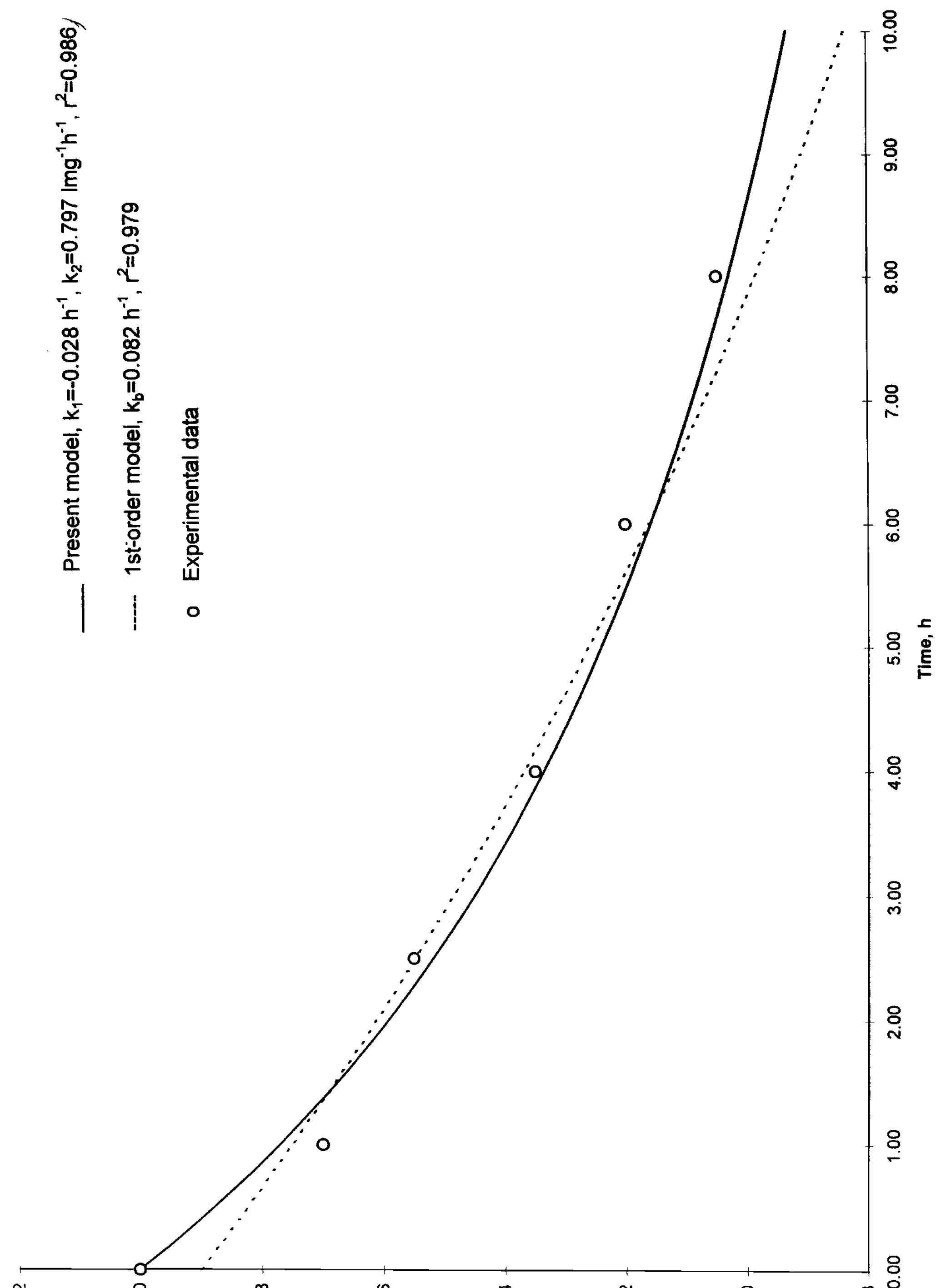

Fig. 6. Comparisons of model predictions and experimental data for tap water.

Table 5. Results for tap water samples

Sampling date	C_0 (mg l^{-1})	Present model			First-order model	
		k_1 (h^{-1})	k_2 (l mg^{-1} h^{-1})	r^2	k_b (h^{-1})	r^2
11/02/97	0.12	0.065	0.913	0.988	0.130	0.984
18/02/97	0.16	−0.015	0.941	0.989	0.086	0.980
25/02/97	0.11	0.076	0.997	0.983	0.147	0.982
04/03/97	0.20	−0.028	0.797	0.986	0.082	0.979

Frankley WTW (fresh samples without re-chlorination) at a fixed temperature 9°C. Once again, the present model prediction is closer to the experimental data than the first-order model prediction. Table 5 shows the results for tap water samples taken on different days. Again, the decay constant, k_2, was found to be slightly different for these samples, which implies that the "reactants" involved in the samples were different. This indicates that the decay constants, k_1 and k_2, determined experimentally with a given water may only be applied to water with the same reactants. For water containing different reactants (TOC), they will have to be determined separately. A comparison of Tables 4 and 5 shows that the values for the rate constants derived for the tap water (Table 5) were, in general, higher than those for the freshly treated water from Frankley WTW (Table 4). There could be several reasons for this, but one possible cause could be the changing balance of reactants with time as a result of multi-stage oxidation processes.

CONCLUSIONS

For Birmingham water, at chlorine concentrations typical of those found in the distribution system, it can be said that:

1. Chlorine does not decay as rapidly in samples which have been rechlorinated as it does in ambient samples. At the same temperature, the re-chlorinated decay constants were 3–10 times lower.
2. The initial concentration has a significant influence on the chlorine decay. The results show that the decay constant was proportional to the reciprocal of the initial concentration.
3. The decay constant is a function of temperature and the data suggest that there is an exponential relationship between the temperature and the product of the initial chlorine concentration and the decay rate. The precise format of the relationship would appear to depend on the concentration and reactivity of the organics in the water.
4. A semi-empirical combined first and second-order model was shown to provide a better description of chlorine decay than the conventional first-order equation.
5. The present model has explained why the chlorine decay occurs more rapidly in the initial stages and why the decay constant of the first-order model decreased with the increase in the initial chlorine concentration.

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APPENDIX A

Development of an Overall Reaction Kinetic Model

Consider a treated water in which there are $\boldsymbol{n}$ chemical components which may react with chlorine. If we assume that the reaction rates of these components with chlorine are second order, then the decay of chlorine can be written as follows:

$$\frac{dC}{dt} = -\lambda_1[X_1]C - \lambda_2[X_2]C - \ldots - \lambda_n[X_n]C \quad (A.1)$$

where $\boldsymbol{C}$ is the concentration of chlorine, $\boldsymbol{t}$ is the time, $[X_i]$ is the concentration of the $\boldsymbol{i}$th chemical component and $\boldsymbol{\lambda_i}$ is its reaction rate constant ($\boldsymbol{i}$ = 1, 2, ..., $\boldsymbol{n}$).

If the number of the chemical components, their reaction rates and their concentrations at one point in time are known, then the concentration of the chlorine at any time can be predicted by equation (A.1). In a real case, however, it is difficult to know how many chemical components are involved in the water and how they react with chlorine. Thus, a possible approach is to establish the reaction equation based on overall reaction stoichiometry. If [X] is the total concentration of the $\boldsymbol{n}$ chemical components and $\boldsymbol{k_o}$ the overall reaction rate constant, then:

$$\frac{dC}{dt} = -k_o C[X] \quad (A.2)$$

In order to find the relationship between X and $\boldsymbol{C}$, the following overall reaction equation is assumed:

$$Cl_2 + \alpha X \Longrightarrow \text{Products} \quad (A.3)$$

where Cl_2 and X represent the chlorine and all the reactants which react with chlorine, respectively, and α is the stoichiometry constant.

In general, α is a function of time as the chemical reaction equations of each reactant may be different. However, as we assume that the reaction rate is established on the basis of overall reaction stoichiometry, α may be regarded as constant. This assumption has been made previously (Dossier-Berne *et al*., 1997).

If $\boldsymbol{C_0}$ and $[X_0]$ are taken to be the initial concentrations of free chlorine and reactants, according to equation A.3, the consumed free chlorine and reactants at time $\boldsymbol{t}$, $\boldsymbol{\Delta C}$ and $\boldsymbol{\Delta}$[X], have the following relationship:

$$\Delta C = C_0 - C = \frac{1}{\alpha}\Delta[X] = \frac{1}{\alpha}([X_0] - [X]) \quad (A.4)$$

Solving [X] from equation (A.4) and substituting it into equation (A.2), yields:

$$\frac{dC}{dt} = -k_o C(\beta + \alpha C) \quad (A.5)$$

where $\boldsymbol{\beta} = ([X_0] - \boldsymbol{\alpha C_0})$ and is a constant related only to the initial concentration of chlorine and the reactants.

Equation (A.5) indicates that the chlorine decay has both first and second-order components. It is interesting to examine some features of equation (A.5).

Consider the case where the initial concentration of the reactants is much higher than that of the chlorine, i.e. $\boldsymbol{\beta} = ([X_0] - \boldsymbol{\alpha C_0}) \gg \boldsymbol{\alpha C}$. With this assumption, the second term in the right-hand side of equation (A.5) is negligible and it can be simplified as:

$$\frac{dC}{dt} = -k_o \beta C = -k_1 C \quad (A.6)$$

Note that $\boldsymbol{\beta}$, and thus $\boldsymbol{k_1}$, decreases with the increase of the chlorine initial concentration. This is why, when the first-order decay model is used, the decay constant often decreases with increased chlorine initial concentration (Dharmarajah and Patania, 1991; Zhang *et al*, 1992). As equation (A.6) is derived on the assumption that the initial chlorine concentration is much smaller than that of the reactants (Chambers *et al*, 1995), it explains why the first-order decay equation fits experimental data very well for lower initial chlorine concentrations and why the regression error of the first-order model with experimental data increases with initial chlorine concentration (Dharmarajah and Patania, 1991). Also, for a given initial concentration, because $\boldsymbol{C}$, and thus $(\boldsymbol{\beta} + \boldsymbol{\alpha C})$, decreases with time, the decay of the chlorine described by equation (A.5) is more rapid initially than later, which is consistent with those data often observed in experiments (Noack and Doerr, 1978; Dharmarajah and Patania, 1991; Jadas-Hécart *et al*, 1992; Zhang *et al*, 1992; Dossier-Berne *et al*, 1997).